

GAURAV KUMAR

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| LinkedIn

SUMMARY

ML Engineer with hands-on experience in adversarial robustness, computer vision, and NLP systems. Pursuing M.Tech in AI/ML with a focus on uncertainty estimation and model reliability. Author of 3 IEEE/IJRASET peer-reviewed publications. Passionate about building production-grade ML pipelines for real-world problems.

EDUCATION

Manav Rachna International Institute of Research and Studies

M.Tech, CSE (AI/ML) | CGPA: 7.56/10 (1st Sem)

Faridabad, India

2024 – Present

Chandigarh University

B.E., Computer Science Engineering | CGPA: 7.20/10

Chandigarh, India

2019 – 2023

TECHNICAL SKILLS

Languages: Python, SQL, Shell Scripting

Tools: Git, Linux, Virtualenv, Jupyter, MLflow

ML/AI: Machine Learning, Deep Learning, Adversarial Robustness, Uncertainty Estimation, Computer Vision, NLP

Libraries: OpenCV, NumPy, Pandas, Scikit-learn, PyTorch, HuggingFace Transformers, spaCy

PROJECTS

Robust Object Detection with Uncertainty-Based Attack Detection | *PyTorch, OpenCV* | [GitHub](#) 2024

- Designed an object detection pipeline with Monte Carlo Dropout uncertainty estimation to flag adversarial and out-of-distribution inputs at inference time.
- Benchmarked against FGSM and PGD attacks; demonstrated measurable robustness improvement over baseline models.

Customer Review Sentiment & Aspect Extraction Pipeline | *BERT, HuggingFace, MLflow* | [GitHub](#) 2025

- Fine-tuned BERT for aspect-level sentiment analysis on Amazon product reviews, extracting structured (aspect, opinion, sentiment) triplets across 8 product categories.
- Automated end-to-end preprocessing, training, and evaluation with MLflow experiment tracking.

Inventory Demand Forecasting with Uncertainty Quantification | *LightGBM, XGBoost, LSTM* | [GitHub](#) 2025

- Built a time-series demand forecasting system with conformal prediction intervals on the M5 Forecasting dataset; compared LightGBM, XGBoost, and LSTM baselines.
- Integrated anomaly detection to suppress irregular demand spikes prior to model training.

AI-Based Fingerprint Enhancement System | *Python, OpenCV* | [GitHub](#) 2023

- Built a preprocessing pipeline using Gabor filtering and morphological ops; improved matching accuracy by ~18% on a 500+ sample benchmark.

EXPERIENCE

Ubuntu LoCo Team

Remote

Open-Source Linux Contributor

2023 – 2025

- Resolved system-level bugs and hardware compatibility issues; contributed patches and documentation across Ubuntu releases.

GirlScript Summer of Code

Remote

Open-Source Contributor — Sound Scape Repository

Summer 2024

- Implemented feature modules for an open-source audio processing project; reviewed peer pull requests under mentorship.

PUBLICATIONS

Securing IoT-Enabled Web Applications & Women's Safety Through Advanced Technologies — IEEE, 2024.

Exploring ApexWolf: A Study of a Cryptocurrency Exchange Website — IEEE, 2024.

Techniques of 2D to 3D Image Reconstruction — IJRASET.

CERTIFICATIONS

Deep Learning & Reinforcement Learning — IBM (Coursera) **Automate Cybersecurity Tasks with Python** — Google (Coursera)